

Why Do Models Underperform on Elusive Objects?

Sohyeon Jang*, Nour El Madhoun[†], Hyunhee Park^{†‡}

*Dept. Applied Artificial Intelligence, Myongji University, Yongin-si, South Korea

[†]LISITE Laboratory, Isep, 10 Rue de Vanves, 92130, Issy-les-Moulineaux, France

[‡]Dept. Information and Communication Engineering, Myongji University, Yongin-si, South Korea

Email: jangso6593@mju.ac.kr; nour.el-madhoun@isep.fr; hhpark@mju.ac.kr

Abstract—Elusive objects are visually similar to their surrounding background, which may encourage models to favor easier background cues over object-specific features that are more difficult to learn. While previous studies have primarily focused on improving detection performance, relatively little attention has been paid to why elusive-object detection fails and whether reliance on background cues contributes to these failures. We propose a diagnostic framework that categorizes samples into background-distracting (BD) and background-neutral (BN) groups based on changes in IoU after suppressing background-region activations. We operationally refer to the response pattern of BD samples as a background shortcut response. We further examine whether background shortcut responses are more prevalent in elusive-object models, whether the observed responses are accompanied by feature-level activation changes, and how they relate to prediction performance. We then extend the analysis to the attribute level by separately evaluating prediction difficulty and background-suppression gains.

Experimental results show that elusive-trained models consistently exhibit a higher proportion of BD samples than salient-trained models. Foreground-region activation changes vary across backbones and feature depths, with BD samples showing larger positive changes than BN samples at several depths. Across all experimental settings, BD samples exhibit lower raw prediction performance than BN samples. Attribute-level analysis further identifies small objects (SO) and indefinable boundaries (IB) as showing consistently high background-suppression gains. This pattern is observed across ResNet-50 and Swin-v2-t when prediction difficulty and background-suppression gains are analyzed as separate dimensions.

Index Terms—elusive object detection, camouflaged object detection, background shortcut learning, region-wise feature ablation, feature intervention

I. INTRODUCTION

The problem of detecting objects whose colors, textures, and boundaries are visually similar to their surrounding background has conventionally been studied under the term camouflaged object detection (COD) [1]. This visual similarity reduces the distinctiveness of object-intrinsic cues and makes accurate detection particularly challenging [6], [7]. In this paper, we use the term *elusive object* to describe this broader phenomenon without restricting it to the specific context of camouflage. By contrast, salient objects are more visually distinguishable from their surrounding background [14].

Deep neural networks tend to rely on predictive cues that are easier to learn than the target concept itself, a tendency commonly described as shortcut learning [8]. Such reliance may improve performance within the training distribution while limiting generalization to unseen environments [9]. For

elusive objects, object-intrinsic visual cues may be particularly weak or ambiguous because of their similarity to the surrounding background [6]. These observations motivate examining whether trained models depend on background cues when object-intrinsic cues are weak. However, previous research has primarily focused on improving detection performance [1]–[5]. Although factors affecting model behavior have also been examined [10], the relationship between background shortcut responses and poor detection performance remains underexplored.

To better understand performance limitations in elusive object detection, we propose a diagnostic framework for quantitatively analyzing background shortcut responses. The framework classifies samples into background-distracting (BD) and background-neutral (BN) groups based on prediction changes before and after background suppression. Using this framework, we address the following research questions:

- RQ1: Are background shortcut responses more prevalent in elusive-object models than in salient-object models?
- RQ2: How do foreground-region activation magnitudes change across feature depths after background suppression?
- RQ3: Are background-distracting samples associated with lower raw prediction performance?
- RQ4: How are elusive-object attributes related to estimated detection difficulty and background shortcut gains?

II. RELATED WORK

A. Camouflaged Object Detection

Camouflaged object detection (COD) aims to segment objects whose visual appearance, including color, texture, and boundary patterns, closely resembles the surrounding environment [1], [2]. The introduction of COD10K established a large-scale benchmark with diverse object categories and rich annotations [2]. Subsequent work introduced NC4K as a test set containing 4,121 images for evaluating model generalization [3], while also proposing camouflaged-object localization and detectability-based ranking tasks.

Representative COD methods have improved segmentation performance by incorporating boundary cues for more accurate contour prediction [4] and by exploiting environmental information in unsupervised learning [5]. These approaches are primarily designed to improve prediction performance rather than to quantify how foreground and background features contribute to model predictions. Consequently, background

dependence in trained COD models remains insufficiently characterized.

B. Shortcut Learning and Context Bias

Shortcut learning refers to the tendency of models to rely on cues that are easier to learn but do not capture the intended task solution [8]. In visual recognition, contextual and background information can act as shortcut cues when predictions depend on correlations with the surrounding context rather than object-specific evidence [9]. Related debiasing research has further shown that such spurious correlations can be learned preferentially when they are easier to acquire than the desired target knowledge [8].

These findings motivate examining background dependence as a potential shortcut, but they do not imply that contextual information is inherently harmful. This distinction is particularly important in COD, where the visual similarity between objects and their surroundings can make background context informative. However, the cited shortcut-learning and debiasing studies do not directly characterize how COD models respond when background features are selectively suppressed. This motivates analyzing both prediction changes and feature responses under controlled background intervention.

C. Feature Attribution and Intervention Analysis

Post-hoc attribution methods are widely used to examine which image regions are associated with model predictions. Grad-CAM generates a coarse localization map by weighting feature maps using target-specific gradients [11], while Grad-CAM++ refines this weighting to improve object localization and account for multiple instances [12]. Although these methods provide spatial explanations of model predictions, they do not directly quantify how suppressing a specified foreground or background feature region changes the segmentation output.

Intervention-based attribution methods instead estimate feature importance from prediction changes caused by modifying internal activations. For example, Ablation-CAM measures changes in the target-class score after ablating individual feature maps and uses these changes to generate a localization map [13]. Its primary objective nevertheless remains visual attribution based on feature-map importance.

Motivated by this intervention principle, our study separately suppresses foreground and background regions in intermediate feature maps and measures the resulting changes in both segmentation predictions and downstream foreground-region activation magnitude. This enables a quantitative analysis of background-dependent responses in COD rather than relying solely on heatmap visualization.

III. DIAGNOSTIC FRAMEWORK

A. Experimental Setup

Datasets. For RQ1–RQ3, COD10K [2] and DUTS [14] provide the training and evaluation data for the elusive-object and salient-object models, respectively. For RQ4, the models are trained on NC4K [3] and evaluated on all 5,066 COD10K images because COD10K is the only dataset used in this study that provides the attribute annotations required for the

attribute-level analysis. For each training dataset, the samples are split into training and validation sets at an 80:20 ratio.

Models. To assess the diagnostic framework across convolutional and Transformer-based backbones, we employ ResNet-50 [15] and Swin-v2-t [16], respectively. Both backbones are initialized with ImageNet-1K-pretrained weights. To isolate the effect of backbone architecture, we integrate each backbone into the same segmentation framework, comprising a U-Net-style decoder and a 1×1 convolutional prediction head.

Training. Each model was optimized for 100 epochs using Adam with a learning rate of 1×10^{-4} and a batch size of 8. The training objective combines binary cross-entropy with logits and soft IoU loss. The checkpoint with the highest validation mIoU was used for all subsequent analyses. During inference, the predicted probability maps are binarized using a threshold of 0.5.

Implementation Details. All experiments were implemented in PyTorch and conducted on a single NVIDIA GeForce GTX 1660 SUPER GPU. Images and masks were resized to 352×352 pixels using bilinear and nearest-neighbor interpolation, respectively, and the images were normalized using ImageNet statistics. Each configuration was trained using three random seeds (0, 1, and 2) under deterministic cuDNN settings. For each training dataset, the same train-validation partition, generated using a fixed split seed of 42, was used across all backbones and seed runs. No additional data augmentation was applied.

B. Region-wise Feature Ablation

Attribution maps show where the model attends, but not how prediction changes after suppressing background features. We therefore employ region-wise feature ablation to quantify prediction quality changes after selectively suppressing foreground- and background-region activations.

To enable localized suppression, the intervention point should preserve spatial correspondence between feature locations and foreground and background regions. Each spatial location aggregates information from a broad image region and may therefore contain both foreground and background information. Consequently, suppressing locations corresponding to the background does not guarantee that only background information is suppressed. In contrast, shallow feature maps retain stronger spatial correspondence with the ground-truth mask, allowing more localized suppression of foreground and background regions. We therefore select a shallow feature stage as the intervention point.

Specifically, the output of conv2_3 (the final block of layer1) is used as the intervention point for ResNet, whereas the output of stage1 is used for Swin. Given an input image I and a binary ground-truth mask M , where $M = 1$ denotes foreground regions and $M = 0$ denotes background regions, the mask is resized to the spatial resolution of the selected feature map using nearest-neighbor interpolation and represented as $\tilde{M}$. In each suppression pass, activations within the target region are set to zero across all channels at the intervention point using $\tilde{M}$.

For the same input image I , three forward passes are performed. The raw forward pass retains the original feature activations, yielding P_{raw} . The foreground-suppressed forward pass sets activations at foreground regions of the selected shallow feature map to zero and produces $P_{\text{fg-sup}}$. Similarly, the background-suppressed forward pass sets activations at background regions of the same feature map to zero and produces $P_{\text{bg-sup}}$.

Unlike input masking, this intervention leaves the input image unchanged. Zeroing is applied only at the intervention point. Since information from neighboring spatial locations is aggregated in subsequent layers, activations at the initially suppressed locations can become nonzero again. Accordingly, we interpret this operation as localized feature suppression rather than the complete removal of foreground or background information from the network. To quantify the prediction quality changes induced by feature suppression, we measure the IoU difference between the raw prediction and each suppressed prediction:

$$\Delta_{\text{IoU}_{bg}} = \text{IoU}_{\text{bg-sup}} - \text{IoU}_{\text{raw}} \quad (1)$$

$$\Delta_{\text{IoU}_{fg}} = \text{IoU}_{\text{raw}} - \text{IoU}_{\text{fg-sup}} \quad (2)$$

Under Eq. (1), a positive value of $\Delta_{\text{IoU}_{bg}}$ suggests that background-region activations may have adversely affected the original prediction, whereas a negative value suggests that they contributed beneficially. Under Eq. (2), a positive value of $\Delta_{\text{IoU}_{fg}}$ suggests that foreground-region activations contributed beneficially to the original prediction, whereas a negative value suggests that they may have adversely affected it. These directional changes serve as the basis for classifying background-suppression responses in the following subsection.

C. Classification of Background Shortcut Responses

Before classifying background-suppression responses, we exclude samples with $\Delta_{\text{IoU}_{fg}} < 0$, for which foreground-region suppression improves prediction. For these samples, the observed behavior cannot be attributed specifically to background-associated interference. We therefore restrict the subsequent analyses to samples satisfying $\Delta_{\text{IoU}_{fg}} \geq 0$. The retained samples are classified according to the sign of $\Delta_{\text{IoU}_{bg}}$ as background-distracting (BD) or background-neutral (BN). BD is defined as

$$\text{BD} = \{\Delta_{\text{IoU}_{bg}} > 0, \Delta_{\text{IoU}_{fg}} \geq 0\} \quad (3)$$

BD comprises samples whose IoU improves after background-region suppression, a pattern consistent with background-associated interference. We operationally refer to this pattern as a background shortcut response. BN is defined as

$$\text{BN} = \{\Delta_{\text{IoU}_{bg}} \leq 0, \Delta_{\text{IoU}_{fg}} \geq 0\} \quad (4)$$

BN comprises the remaining samples whose IoU does not improve after background-region suppression. This category includes cases in which suppression decreases IoU and cases in which IoU remains unchanged. Thus, BN serves as the complementary category to BD and does not imply the absence of background influence.

D. Foreground-Region Activation Changes under Background Suppression

The IoU increase observed for BD samples does not necessarily indicate an increase in foreground-region activation magnitude. It may instead result from the object becoming more distinguishable in the prediction after background-region suppression. To determine whether foreground-region activation magnitude also increases, we compare this quantity between the raw and background-suppressed passes. We compute the root mean square (RMS) over all activations across channels at spatial locations within the resized foreground mask. For a feature map F , the foreground-region RMS is defined as

$$\text{RMS}_{\text{fg}}(F) = \sqrt{\frac{1}{N_{\text{fg}}} \sum_{k=1}^{N_{\text{fg}}} a_k^2} \quad (5)$$

where a_k denotes an activation within the foreground region, and N_{fg} denotes the total number of foreground-region activations across spatial locations and channels.

For the same input and feature stage, let F_{raw} and $F_{\text{bg-sup}}$ denote the feature maps obtained from the raw and background-suppressed passes, respectively. Their foreground-region RMS values are defined as

$$R_{\text{fg}}^{\text{raw}} = \text{RMS}_{\text{fg}}(F_{\text{raw}}) \quad (6)$$

$$R_{\text{fg}}^{\text{bg-sup}} = \text{RMS}_{\text{fg}}(F_{\text{bg-sup}}) \quad (7)$$

The change in foreground activation magnitude induced by background suppression, ΔR_{fg} , is defined as

$$\Delta R_{fg} = R_{\text{fg}}^{\text{bg-sup}} - R_{\text{fg}}^{\text{raw}} \quad (8)$$

We compute ΔR_{fg} separately at layer1–layer4 of ResNet and stage1–stage4 of Swin to assess foreground-region activation changes at each feature stage after background suppression. A positive value of ΔR_{fg} indicates increased foreground-region activation magnitude, whereas values close to zero indicate little effect of background suppression on foreground-region activation magnitude.

E. Prediction Performance Comparison between BD and BN

To examine whether BD samples exhibit lower raw prediction performance than BN samples, we compare their raw predictions. Test samples are classified as BD or BN using the criteria defined above. We report the group-averaged IoU, weighted F-measure (F_w) [17], and mean absolute error (MAE). IoU measures region overlap between the predicted mask and the ground-truth mask. Weighted F-measure evaluates foreground-map quality by weighting prediction errors according to their spatial location and neighborhood structure. MAE measures the average absolute difference between the prediction map and the ground-truth mask.

F. Attribute-Level Analysis of Detection Difficulty and Background Shortcut Responses

The BD/BN classification identifies which samples exhibit background shortcut responses but does not reveal which object attributes are associated with these responses. To examine these associations, we use the COD10K annotations

TABLE I
BD AND BN SAMPLE COUNTS AND BD RATIOS REPORTED AS MEAN $\pm$ SD.

Datatype	Model	Split	N	N_{BN}	N_{BD}	BD Ratio
Elusive	ResNet	Train	3137 $\pm$ 23	2061 $\pm$ 176	1076 $\pm$ 180	34.3% $\pm$ 5.7%
Elusive	ResNet	Test	1845 $\pm$ 12	617 $\pm$ 99	1228 $\pm$ 91	66.6% $\pm$ 5.2%
Salient	ResNet	Train	3192 $\pm$ 35	2731 $\pm$ 284	461 $\pm$ 281	14.5% $\pm$ 8.8%
Salient	ResNet	Test	1963 $\pm$ 54	1030 $\pm$ 160	933 $\pm$ 170	47.5% $\pm$ 8.2%
Elusive	Swin	Train	3131 $\pm$ 31	1392 $\pm$ 116	1738 $\pm$ 139	55.5% $\pm$ 4.0%
Elusive	Swin	Test	1746 $\pm$ 119	391 $\pm$ 27	1354 $\pm$ 97	77.6% $\pm$ 0.8%
Salient	Swin	Train	3211 $\pm$ 12	2853 $\pm$ 126	358 $\pm$ 117	11.1% $\pm$ 3.7%
Salient	Swin	Test	1902 $\pm$ 35	1002 $\pm$ 28	900 $\pm$ 50	47.3% $\pm$ 1.9%

$N = N_{BN} + N_{BD}$, N_{BN} : number of BN samples, N_{BD} : number of BD samples.

for small object (SO), indefinable boundary (IB), occlusion (OC), multiple objects (MO), shape complexity (SC), and out-of-view (OV). Big object (BO) is excluded because its limited sample size does not support reliable attribute-level comparison.

An attribute may be associated with both detection difficulty and background-suppression gains. This overlap makes either measure alone insufficient for attribute-level interpretation. Analysis 1 jointly maps estimated detection difficulty and mean background-suppression gain to compare attributes across these two dimensions. Interpreting the gain dimension requires accounting for raw IoU, which constrains the possible range of ΔIoU_{bg} . Analysis 2 mitigates this limitation by examining whether attribute-wise gain differences persist within raw-IoU strata.

Analysis 1: Attribute-Level Mapping of Detection Difficulty and Background-Suppression Gains.

To estimate the association of each attribute with detection difficulty while accounting for the other included attributes, we fit a multiple linear regression model. The dependent variable is IoU_{raw} , and the binary independent variables indicate the presence or absence of each attribute:

$$\text{IoU}_{\text{raw}} = \beta_0 + \sum_{k \in \mathcal{A}} \beta_k x_k + \epsilon \quad (9)$$

where $\mathcal{A} = \{\text{SO}, \text{IB}, \text{OC}, \text{MO}, \text{SC}, \text{OV}\}$ and $x_k \in \{0, 1\}$ indicates whether attribute k is present. The coefficients are estimated using ordinary least squares. A negative $\hat{\beta}_k$ indicates that attribute k is associated with lower raw prediction performance after accounting for the other included attributes. We define the attribute-level difficulty score as $d_k = -\hat{\beta}_k$. Positive d_k values indicate associations with lower raw IoU, with higher scores corresponding to larger estimated associations. Negative d_k values indicate associations with higher raw IoU.

Next, we compute the mean background-suppression gain for each attribute:

$$g_k = \frac{1}{N_k} \sum_{i \in \mathcal{I}_k} \Delta\text{IoU}_{bg,i}, \quad (10)$$

Here, $\mathcal{I}_k$ denotes the set of analyzed images containing attribute k , and $N_k = |\mathcal{I}_k|$. Higher g_k values indicate greater mean background-suppression gains. Because images may contain multiple attributes, g_k is an attribute-wise descriptive mean rather than an estimate of an independent attribute effect.

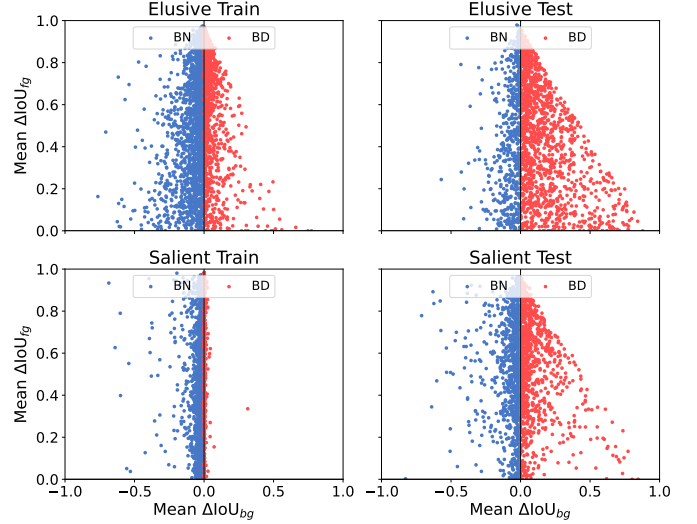


Fig. 1. BD/BN distributions for ResNet based on mean ΔIoU_{bg} and ΔIoU_{fg} .

For each attribute, we plot (g_k, d_k) to compare attributes along the background-suppression-gain and detection-difficulty dimensions. The medians of g_k and d_k serve as relative reference lines rather than absolute thresholds.

Analysis 2: Raw-IoU-Stratified Analysis of Background-Suppression Gains.

Attribute-level mean gains may be influenced by baseline raw IoU because samples with lower raw IoU have a wider possible range of improvement after suppression. We therefore group the analyzed images into raw-IoU bins and compare attribute-wise mean ΔIoU_{bg} values within each bin. This stratified comparison evaluates whether an attribute exhibits relatively large background-suppression gains among samples with similar baseline performance.

IV. EXPERIMENTS

A. RQ1: Prevalence of Background Shortcut Responses

RQ1 examines whether background shortcut responses are more prevalent in elusive-object models than in salient-object models. Using the operational criteria defined in III-C, we compute the BD ratio for each seed as $N_{BD}/(N_{BD} + N_{BN})$, where the denominator includes only samples satisfying $\Delta\text{IoU}_{fg} \geq 0$. Fig. 1 qualitatively illustrates the higher prevalence of BD samples in the elusive-object ResNet models than

TABLE II
RAW-PREDICTION METRICS FOR BD AND BN SAMPLES, REPORTED AS MEAN $\pm$ SD.

Datatype	Model	Group	N	IoU $\uparrow$	F_ω $\uparrow$	MAE $\downarrow$
Elusive	ResNet	BN	617 $\pm$ 99	0.712 $\pm$ 0.036	0.774 $\pm$ 0.045	0.026 $\pm$ 0.006
Elusive	ResNet	BD	1228 $\pm$ 91	0.539 $\pm$ 0.012	0.611 $\pm$ 0.007	0.050 $\pm$ 0.005
Salient	ResNet	BN	1030 $\pm$ 160	0.870 $\pm$ 0.003	0.915 $\pm$ 0.004	0.020 $\pm$ 0.002
Salient	ResNet	BD	933 $\pm$ 170	0.778 $\pm$ 0.015	0.830 $\pm$ 0.014	0.041 $\pm$ 0.003
Elusive	Swin	BN	391 $\pm$ 27	0.746 $\pm$ 0.009	0.820 $\pm$ 0.010	0.032 $\pm$ 0.003
Elusive	Swin	BD	1354 $\pm$ 97	0.593 $\pm$ 0.002	0.665 $\pm$ 0.002	0.037 $\pm$ 0.002
Salient	Swin	BN	1002 $\pm$ 28	0.898 $\pm$ 0.003	0.937 $\pm$ 0.003	0.017 $\pm$ 0.000
Salient	Swin	BD	900 $\pm$ 50	0.785 $\pm$ 0.004	0.835 $\pm$ 0.003	0.036 $\pm$ 0.002

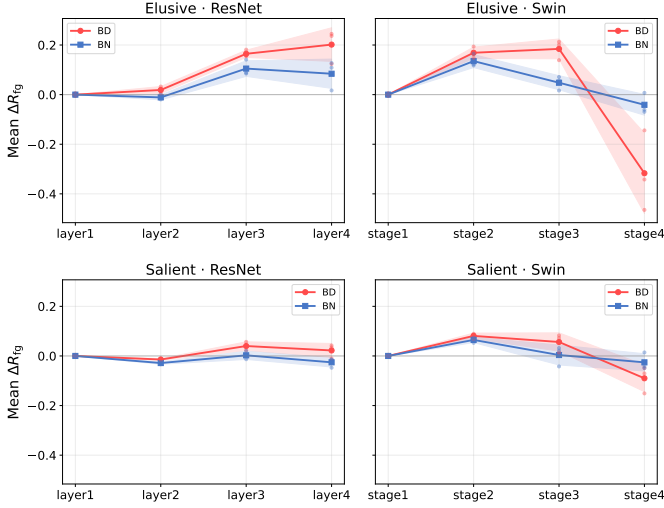


Fig. 2. Depth-wise ΔR_{fg} for BD and BN samples, with lines showing the mean and shaded regions indicating $\pm$ SD.

in the corresponding salient-object models, consistent with Table I.

As shown in Table I, elusive-object models exhibited higher BD ratios than salient-object models for both backbones across both data splits. On the respective training splits, the BD ratios of the elusive-object models exceeded those of the salient-object models by 19.8 percentage points for ResNet and 44.4 percentage points for Swin. On the respective test splits, the corresponding gaps were 19.1 and 30.3 percentage points.

Overall, background shortcut responses were consistently more prevalent in elusive-object models than in salient-object models across both backbones and both data splits. However, the IoU improvement observed in BD samples may reflect greater object distinguishability after background suppression without necessarily indicating an increase in foreground-region activation magnitude. RQ2 examines whether foreground-region activation magnitude also increases after background suppression.

B. RQ2: Depth-Wise Foreground-Region Activation Changes after Background Suppression

RQ2 examines foreground-region activation changes after background suppression using ΔR_{fg} defined in Section III-D. We measure ΔR_{fg} at layer1–layer4 of ResNet and stage1–stage4 of Swin on the respective test splits. Fig. 2 presents the resulting depth-wise profiles. At the intervention depth,

$\Delta R_{fg} = 0$ by design because background suppression does not directly modify foreground-region activations at that point.

For the elusive-trained ResNet, mean ΔR_{fg} became positive in the deeper layers. The BD mean increased through layer4, whereas the BN mean peaked at layer3 and decreased slightly at layer4. Consequently, the descriptive BD–BN gaps were larger at layer3 and layer4. For the elusive-trained Swin, both groups showed positive mean changes at the intermediate stages, and the largest BD–BN mean gap appeared at stage3. At stage4, however, the mean responses became negative, particularly for BD samples, with marked variation across seeds. The positive foreground-region activation changes observed at the intermediate Swin stages thus did not extend to the terminal stage. In the salient-trained models, the changes were generally smaller and the BD and BN profiles were closer, although both groups in Swin again showed negative mean responses at stage4.

Overall, the depth-wise profiles show that foreground-region activation changes after background suppression varied across backbones and feature depths. At depths with positive mean responses, BD samples generally showed larger mean ΔR_{fg} than BN samples, particularly in the elusive-trained models. The prediction improvement defining BD responses can coincide with a larger positive shift in foreground-region activation magnitude. This descriptive association does not establish that the activation change causes the IoU improvement, and RMS magnitude alone does not measure the recovery or quality of foreground representations. Identifying the architectural mechanisms underlying the negative stage4 responses in Swin is beyond the scope of this analysis.

C. RQ3: Prediction Performance of BD and BN Samples Before Background Suppression

RQ3 examines whether BD samples exhibit lower segmentation performance than BN samples before background suppression. Table II summarizes the mean values of IoU, weighted F_ω , and MAE for BD and BN samples on the respective test splits.

For every evaluated model, BD samples exhibited lower IoU and F_ω values and higher MAE values than BN samples. For IoU, the BN–BD gaps were 0.17 and 0.09 for the elusive- and salient-object ResNet models, respectively, and 0.15 and 0.11 for the corresponding Swin models. The F_ω gaps followed the same pattern, with 0.16 and 0.09 for the ResNet models and 0.16 and 0.10 for the Swin models, respectively. Although

TABLE III
COMPUTATIONAL OVERHEAD IN MS/IMAGE, WITH TIMING VALUES REPORTED AS MEAN $\pm$ SD.

Datatype	Model	Raw	BG-sup	FG-sup	Total
Elusive	ResNet	24.20 $\pm$ 0.13	24.27 $\pm$ 0.13	24.28 $\pm$ 0.07	72.75 $\pm$ 0.33
Salient	ResNet	25.10 $\pm$ 1.40	25.09 $\pm$ 1.38	25.15 $\pm$ 1.36	75.35 $\pm$ 4.14
Elusive	Swin	20.70 $\pm$ 0.40	20.74 $\pm$ 0.38	20.80 $\pm$ 0.44	62.24 $\pm$ 1.22
Salient	Swin	21.91 $\pm$ 1.66	21.92 $\pm$ 1.67	21.95 $\pm$ 1.64	65.79 $\pm$ 4.97

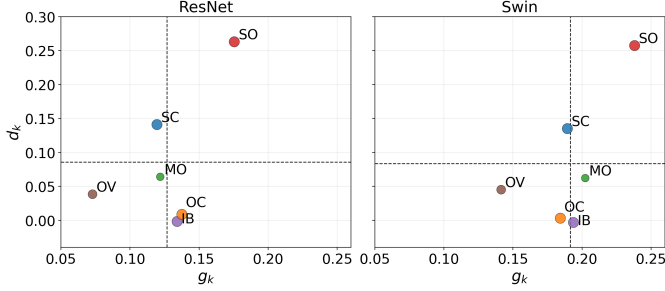


Fig. 3. Estimated Detection Difficulty (d_k) and Mean Background-Suppression Gain (g_k) Across Attributes.

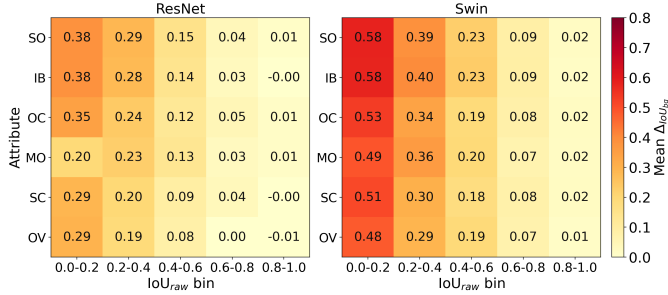


Fig. 4. Heatmaps of Attribute-Wise Mean ΔIoU_{bg} Across Raw-IoU Bins.

MAE was also consistently higher for BD samples, the differences were less pronounced than those observed for IoU and F_w . Taken together, these results indicate that background shortcut responses are associated with lower raw prediction performance across all three metrics.

D. RQ4: Attribute-Level Detection Difficulty and Background-Suppression Gains

RQ4 examines how the six COD10K attributes selected in Section III-F are related to detection difficulty and background shortcut responses. Fig. 3 compares their relative positions along the detection-difficulty (d_k) and mean background-suppression-gain (g_k) dimensions. Because g_k may be influenced by baseline raw IoU, Fig. 4 complements its interpretation by examining whether the relative gain patterns persist within the predefined 0.2-wide raw-IoU bins.

Relative to the median lines in Fig. 3, SO was relatively high on both dimensions, whereas SC was high mainly in detection difficulty and IB was high mainly in background-suppression gain. OV was relatively low on both dimensions, while MO and OC exhibited backbone-dependent patterns.

Fig. 4 compares attribute-wise mean ΔIoU_{bg} within the predefined raw-IoU bins. This bin-wise comparison reduces the potential influence of baseline raw IoU on g_k , as higher base-

line performance leaves less room for improvement. Across the low-to-middle raw-IoU bins, SO and IB consistently showed the largest or near-largest gains for both ResNet and Swin. The strong response of SO observed in Fig. 3 remained evident, while the response of IB became more apparent within comparable baseline-performance ranges.

When the findings from Figs. 3 and 4 are considered jointly, estimated detection difficulty alone is insufficient to characterize attribute-level model behavior in elusive-object detection. Some attributes exhibited relatively strong background shortcut responses despite relatively low estimated detection difficulty, whereas others showed the opposite tendency. These findings suggest that both estimated detection difficulty and background shortcut responses should be considered when characterizing model behavior across elusive-object attributes.

E. Computational Cost of the Diagnostic Framework

To assess computational cost, we measured per-image processing time using the hardware and evaluation settings described in Section III-A, with a batch size of 8. Data loading and host-to-device transfer were excluded, the first five batches were used for warm-up, and CUDA was synchronized immediately before and after each timed operation. Table III reports the mean and standard deviation across the three seed-specific runs. Across the mean values reported in Table III, each feature-suppressed pass differed from the corresponding raw pass by no more than 0.10 ms/image ($< 0.5\%$ of the raw-pass time), indicating negligible per-pass computational overhead. Nevertheless, the complete diagnostic procedure requires one raw pass and two feature-suppressed passes. The total time, computed as the sum of these three pass times, ranged from 62.24 to 75.35 ms/image, approximately three times the measured raw-pass time. Because the proposed framework is intended for post-hoc diagnosis, this additional cost is incurred only during diagnostic analysis and does not modify the model's standard inference path.

V. DISCUSSION: INTEGRATION AND EXTENSION WITH BLOCKCHAIN TECHNOLOGY

The proposed BD/BN diagnostic framework can be integrated with blockchain technology and extended toward a trustworthy elusive object detection ecosystem. The sample-level quantitative indicators produced by the framework (ΔIoU_{bg} , ΔR_{fg} , and the BD ratio) are reproducible by any party given a fixed model checkpoint and ground-truth mask, and are compact in size; as such, they can serve as verifiable evidence suitable for on-chain recording. In other words, the framework goes beyond a tool for diagnosing model

internals and provides a quantitative interface that can be directly connected to blockchain-based trust infrastructures. Specifically, a shortcut-auditing pipeline can be constructed in which the diagnostic results are recorded on a blockchain. In safety-critical domains such as medical lesion segmentation and autonomous driving, anchoring per-release BD ratios and attribute-level shortcut statistics on a permissioned blockchain allows third parties to automatically verify, via smart contracts, whether a deployed model satisfies predefined shortcut-robustness criteria without accessing the model weights. Extending this to federated learning environments, the BD ratio of each client's local data can be recorded on-chain as a data-quality signal, and smart contracts can implement a shortcut-aware incentive mechanism that rewards the contribution of background-neutral (BN-rich) data. Furthermore, managing the provenance of ground-truth masks and attribute annotations (e.g., SO and IB), upon which the diagnosis itself relies, completes an end-to-end trust chain that also guarantees the integrity of the diagnostic results. In this manner, the proposed framework can serve as a key building block of a blockchain-integrated structure spanning diagnosis, anchoring, verification, and incentivization, and we plan to pursue the concrete design and on-chain overhead analysis of this pipeline in future work.

VI. CONCLUSION

This paper proposed a region-wise feature-ablation framework to quantitatively diagnose background shortcut responses in elusive object detection. The framework classifies samples into background-distracting (BD) and background-neutral (BN) groups based on prediction changes after foreground and background suppression.

Experimental results showed that elusive-trained models consistently exhibited higher proportions of BD samples than salient-trained models across both backbones. Compared with BN samples, BD samples generally exhibited larger increases in foreground-region activation magnitude after background suppression at several feature depths. BD samples also showed lower raw prediction performance than BN samples. Together, these findings indicate that background shortcut responses are associated with poorer prediction performance in elusive object detection. At the attribute level, the relationship between background shortcut responses and estimated detection difficulty varied across elusive-object attributes. These findings suggest that both estimated detection difficulty and background shortcut responses should be considered when characterizing performance across elusive-object attributes.

The core diagnostic procedure is not tied to the specific datasets or architectures used in this study. It requires a ground-truth foreground mask and an intermediate feature map that preserves spatial correspondence with the mask. The framework may therefore be extended to other binary segmentation datasets and backbones with suitable spatial feature maps, provided that the intervention point is adapted to each architecture's feature resolution and organization. The proposed framework can be used for post-hoc model auditing

to identify samples vulnerable to background shortcuts and to improve augmentation, sampling, and loss design based on object attributes with high shortcut susceptibility. However, this study focuses on diagnosing background shortcut responses rather than mitigating them. Future work will investigate shortcut-mitigation training using BD samples and reconstructed background-object combinations and evaluate whether new mitigation methods effectively reduce background dependence.

ACKNOWLEDGMENT

This work was supported by the National Research Foundation of Korea (NRF) funded by the Ministry of Education (MSIT) and the Technology Development Program funded by the Ministry of SMEs and Startups (MSS) of the Republic of Korea (No. 2022R1A2C2005705, No. RS-2025-02217793).

REFERENCES

- [1] D.-P. Fan, G.-P. Ji, G. Sun, M.-M. Cheng, J. Shen, and L. Shao, "Camouflaged object detection," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 2774–2784, doi: 10.1109/CVPR42600.2020.00285.
- [2] D.-P. Fan, G.-P. Ji, M.-M. Cheng, and L. Shao, "Concealed object detection," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 10, pp. 6024–6042, 2022, doi: 10.1109/TPAMI.2021.3085766.
- [3] Y. Lv, J. Zhang, Y. Dai, A. Li, B. Liu, N. Barnes, and D.-P. Fan, "Simultaneously localize, segment and rank the camouflaged objects," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 11586–11596.
- [4] Y. Sun, S. Wang, C. Chen, and T.-Z. Xiang, "Boundary-guided camouflaged object detection," in *Proc. 31st International Joint Conference on Artificial Intelligence*, 2022, pp. 1335–1341.
- [5] J. Du, F. Hao, M. Yu, D. Kong, J. Wu, B. Wang, J. Xu, and P. Li, "Shift the lens: Environment-aware unsupervised camouflaged object detection," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 19271–19282, doi: 10.1109/CVPR52734.2025.01795.
- [6] J. Zhong, A. Wang, C. Ren, and J. Wu, "A survey on deep learning-based camouflaged object detection," *Multimedia Systems*, vol. 30, no. 5, Art. no. 268, 2024, doi: 10.1007/s00530-024-01478-7.
- [7] Y. Liang, G. Qin, M. Sun, X. Wang, J. Yan, and Z. Zhang, "A systematic review of image-level camouflaged object detection with deep learning," *Neurocomputing*, vol. 566, Art. no. 127050, 2024, doi: 10.1016/j.neucom.2023.127050.
- [8] J. Nam, H. Cha, S. Ahn, J. Lee, and J. Shin, "Learning from failure: De-biasing classifier from biased classifier," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 20673–20684.
- [9] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, vol. 2, no. 11, pp. 665–673, 2020, doi: 10.1038/s42256-020-00257-z.
- [10] Y. Zhang, J. Zhang, W. Hamidouche, and O. Déforges, "Predictive uncertainty estimation for camouflaged object detection," *IEEE Transactions on Image Processing*, vol. 32, pp. 3580–3591, 2023, doi: 10.1109/TIP.2023.3287137.
- [11] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE International Conference on Computer Vision*, 2017, pp. 618–626, doi: 10.1109/ICCV.2017.74.
- [12] A. Chattopadhyay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks," in *2018 IEEE Winter Conference on Applications of Computer Vision*, 2018, pp. 839–847.
- [13] S. Desai and H. G. Ramaswamy, "Ablation-CAM: Visual explanations for deep convolutional network via gradient-free localization," in *2020 IEEE Winter Conference on Applications of Computer Vision*, 2020, pp. 972–980.

- [14] L. Wang, H. Lu, Y. Wang, M. Feng, D. Wang, B. Yin, and X. Ruan, "Learning to detect salient objects with image-level supervision," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 136–145.
- [15] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 770–778, doi: 10.1109/CVPR.2016.90.
- [16] Z. Liu, H. Hu, Y. Lin, Z. Yao, Z. Xie, Y. Wei, J. Ning, Y. Cao, Z. Zhang, L. Dong, F. Wei, and B. Guo, "Swin Transformer V2: Scaling up capacity and resolution," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 12009–12019.
- [17] R. Margolin, L. Zelnik-Manor, and A. Tal, "How to evaluate foreground maps?" in *Proc. IEEE Conference on Computer Vision and Pattern Recognition*, 2014, pp. 248–255, doi: 10.1109/CVPR.2014.39.